Reviewer Pack — Minimal Hodge Index and Resolution Proof Width Inequality

Entry 70a5276cbba7 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-06-04 20:09:56 UTC

Minimal Hodge Index and Resolution Proof Width Inequality

Entry ID: `70a5276cbba7` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-06-04 20:09:56 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Hodge Theory) **Field B** (complexity object): Resolution Proof Complexity

Statement:

> For every k-CNF formula φ with n variables, the resolution proof width w(φ) satisfies the inequality log2(n^(k+1)) ≤ w(φ) + H(φ), where H(φ) is the minimal Hodge index of φ's associated algebraic variety over the complex numbers.

Rationale (proposer's reasoning):

> The Hodge index measures the complexity of an algebraic variety and is related to the geometric properties of the variety. By connecting this invariant with resolution proof width, we may reveal new insights into the structure of SAT instances and their hardness.

Taxonomy category: `AlgebraicGeometry × ResolutionProofComplexity` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `1189b4cf97616555`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The conjecture is supported if the inequality log2(n^(k+1)) ≤ w(φ) + H(φ) holds for all generated k-CNF formulas φ with n variables, where w(φ) is the resolution proof width and H(φ) is the minimal Hodge index. The conjecture is falsified if there exists any k-CNF formula φ such that log2(n^(k+1)) > w(φ) + H(φ).

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM₁	LLM₂
RELATIVIZATION	UNCERTAIN	1.00	SAFE	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - 'Hodge Theory' AND 'resolution proof complexity' AND 'proof width inequality' - 'algebraic geometry' AND 'minimal Hodge index' AND 'CNF formula resolution' - 'complex algebraic variety' AND 'resolution proof width' AND 'logarithmic inequality'

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.2s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    # Parameters for k-CNF generation
    n = 10  # Number of variables
    m = 2 * n  # Number of clauses (2 literals per clause)
    k = 3  # Clause length

    # Generate a random k-CNF formula φ with n variables and m clauses
    phi = []
    for _ in range(m):
        clause = set(random.sample(range(1, n + 1), k))
        phi.append(clause)

    # Convert the k-CNF to a list of literals
    literals = [0] * (n + 1)
    for clause in phi:
        for literal in clause:
            literals[literal] += 1

    # Calculate the resolution proof width w(φ)
    def dpll(phi):
        if not phi:
            return 0
        unit_clauses = [c for c in phi if len(c) == 1]
        if not unit_clauses:
            return max(dpll([c - {l} for c in phi if l not in c]) for l in literals if literals[l]
        literal, _ = unit_clauses[0]
        new_phi = [c - {literal} for c in phi if literal not in c]
        return 1 + dpll(new_phi)
```

```
    w_phi = dpll(phi)

    # Calculate the minimal Hodge index H(φ) (simplified example)
    H_phi = sum(literals[l] ** 2 for l in range(1, n + 1))

    # Check the inequality log2(n^(k+1)) ≤ w(φ) + H(φ)
    lhs = math.log2(n ** (k + 1))
    rhs = w_phi + H_phi

    conjecture_holds = lhs <= rhs
    counterexample = "" if conjecture_holds else f"lhs={lhs}, rhs={rhs}"

    return {
        "metric_name": "resolution_proof_width",
        "metric_value": w_phi,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29]

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    # Compute mean/std of metric_value
    total_metric_value = sum(r["metric_value"] for r in results)
    mean_metric_value = total_metric_value / len(results)
    variance = sum((r["metric_value"] - mean_metric_value) ** 2 for r in results) / len(results)
    std_metric_value = math.sqrt(variance)

    # Compute fraction of seeds where conjecture_holds
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    # Determine the final result
    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conj
        print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed
    else:
        print("RESULT: INCONCLUSIVE reason=unknown")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
]) for l in literals if literals[l] > 0)
```

```
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f47baa54.py", line 44, in dpll
    return max(dpll([c - {l} for c in phi if l not in c]) for l in literals if literals[l] > 0)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f47baa54.py", line 44, in <genexpr>
    return max(dpll([c - {l} for c in phi if l not in c]) for l in literals if literals[l] > 0)
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f47baa54.py", line 44, in dpll
    return max(dpll([c - {l} for c in phi if l not in c]) for l in literals if literals[l] > 0)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f47baa54.py", line 44, in <genexpr>
    return max(dpll([c - {l} for c in phi if l not in c]) for l in literals if literals[l] > 0)
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f47baa54.py", line 44, in dpll
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           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
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           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
 File "/hom
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> The test crashed before producing any data, which means we cannot verify if the inequality log2(n^(k+1)) ≤ w(φ) + H(φ) holds for all generated k-CNF formulas. Therefore, the verdict is inconclusive. | next: Re-run the test without crashing to verify the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	22934
2	propose	ollama_remote	glm4:latest	0	0	14117
3	preregistration	ollama_remote	glm4:latest	0	0	10101
4	novelty	ollama_remote	glm4:latest	0	0	9125
5	novelty	ollama_remote	glm4:latest	0	0	15189
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16329
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10736
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13368
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9974
10	judge	ollama_remote	glm4:latest	0	0	15672

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 137544 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/70a5276cbba7.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/70a5276cbba7.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/70a5276cbba7.tar.gz` (if generated)